2024

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL - 1902

ANNUAL REPORT ON

SDG-3

GOOD HEALTH AND WELL-BEING

PROGRESS ON SDG-3

[GOOD HEALTH AND WELL BEING]

Mawlana Bhashani Science and Technology University (MBSTU) has actively advanced SDG 3: Good Health and Well-being during 2024 by integrating health-focused efforts across its core functions. Academic programs of the university, particularly in the Faculty of Life Science, directly support health promotion by addressing diverse health issues, including infectious disease, epidemiology, medicine, food safety and environmental impacts on public health. Furthermore, MBSTU engaged in several research projects specifically targeting health-related issues, contributing to real-world solutions and knowledge creation. More than 50 research articles related to this specific SDG were published in Scopus indexed journals during 2024.

On campus, the university provided essential student health services and wellness programs to ensure the well-being of its population. Importantly, MBSTU extended its impact beyond the campus through community outreach activities focused on health. All these efforts, including student services, research, and outreach, demonstrate the commitment of the university to transparency and measuring the practical outcomes of its contribution to promoting healthier lives.

UNIVERSITY COLLABORATIONS FOR HEALTH RESEARCH, POLICY, AND RESILIENCE

MBSTU collaborates with local, national, and global health institutions to advance research, policy, and practice in improving human health and wellbeing. Faculty members engage in joint research projects, capacity-building initiatives, and knowledge exchange with national and international universities, health agencies and hospitals, international research organizations (e.g., icddr, b), and global public health consortia. The collaborative studies (evidenced by co-authored publications) address critical health challenges including infectious diseases, non-communicable diseases, environmental health, and public health resilience. These partnerships enhance innovation, promote evidence-based interventions, and strengthen our collective impact toward achieving Good Health and Wellbeing.

Determinants of immune responses predictive of protection against shigellosis in an endemic zone: a systems analysis of antibody profiles and function

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Summary

Background *Shigella* is the third leading global cause of moderate or severe diarrhoea among children younger than 5 years globally, and is the leading cause in children aged 24–59 months. The mechanism of protection against *Shigella* infection and disease in endemic areas is uncertain. We aimed to compare the *Shigella*-specific antibody responses in individuals living in *Shigella*-endemic and non-endemic areas, and to identify correlates of protection in a *Shigella*-endemic location.

Methods We applied a systems approach to retrospectively analyse serological responses to *Shigella* across endemic and non-endemic populations. We profiled serum samples collected from 44 individuals from the USA without previous exposure to *Shigella* and who were experimentally challenged with *Shigella sonnei* (non-endemic setting), and serum samples collected from 55 Peruvian army recruits (endemic setting). In the endemic setting, a subset of 37 samples collected from individuals infected with culture-confirmed *Shigella flexneri* 2a were divided into two groups: susceptible, which included individuals infected within 90 days of entering the camp (n=29); or resistant, which included individuals infected later than 90 days after entering the camp (n=8). We analysed *Shigella*-specific antibody isotype, subclass, and Fc receptor binding profiles across IpaB, IpaC, IpaD, and lipopolysaccharide from *S flexneri* 2a, 3a, and 6, and *S sonnei*, and O-specific polysaccharide (OSP) from *S flexneri* 2a and 3a and *S sonnei*. We also evaluated antibody-mediated complement deposition and innate immune cell activation. The main outcome of interest was the detection of antibody markers and functionality associated with protection against shigellosis in a high-burden endemic setting.

Findings Adults with endemic exposure to *Shigella* possessed broad and functional antibody responses across polysaccharide, glycolipid, and protein antigens compared with individuals from non-endemic regions. In a setting with high *Shigella* burden, elevated levels of OSP-specific Fcα receptor (FcαR) binding antibodies were associated with resistance to shigellosis, whereas total OSP-specific IgA was not, suggesting a potentially unique functionality. OSP-specific FcαR binding IgA found in resistant individuals activated bactericidal neutrophil functions including phagocytosis, degranulation, and production of reactive oxygen species. Moreover, IgA depletion from resistant serum significantly reduced binding of OSP-specific antibodies to FcαR and antibody-mediated activation of neutrophils and monocytes.

Interpretation Our findings suggest that OSP-specific functional IgA responses contribute to protective immunity against *Shigella* infection in a high-burden setting. These findings will assist in the development and evaluation of *Shigella* vaccines.

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Introduction

Diarrhoeal diseases caused by enteric pathogens are a major cause of death in children younger than 5 years in low-income and middle-income countries.[1,2] Attempts to develop effective vaccines against enteric pathogens have been impeded by both the breadth of pathogens involved, and incomplete understanding of both correlates and mediators of protective immunity against individual pathogens in endemic settings. Vaccines that have been protective against enteric pathogens, including rotavirus,[3] *Shigella*,[4] and *Vibrio cholerae*,[5] in non-endemic settings have provided decreased protection when evaluated in endemic areas. Individuals living in *Shigella*-endemic regions are exposed to *Shigella* infection from childhood and develop partially protective immunity against shigellosis during the first years of life.[6] However, our understanding of the immune response that evolves in these individuals is incomplete. Deciphering correlates of protection against

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Evaluation of 3β-hydroxysteroid dehydrogenase activity using progesterone and androgen receptors-mediated transactivation

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3β-Hydroxysteroid dehydrogenases (3β-HSDs) catalyze the oxidative conversion of delta (5)-ene-3-beta-hydroxy steroids and ketosteroids. Human 3β-HSD type 2 (HSD3B2) is predominantly expressed in gonadal and adrenal steroidogenic cells for producing all classes of active steroid hormones. Mutations in HSD3B2 gene cause a rare form of congenital adrenal hyperplasia with varying degree of salt wasting and incomplete masculinization, resulting from reduced production of corticoids and androgens. Therefore, evaluation of the HSD3B2 enzymatic activity in both pathways for each steroid hormone production is important for accurately understanding and diagnosing this disorder. Using progesterone receptor (PR)- and androgen receptor (AR)-mediated transactivation, we adapted a method that easily evaluates enzymatic activity of HSD3B2 by quantifying the conversion from substrates [pregnenolone (P5) and dehydroepiandrosterone (DHEA)] to (progesterone and androstenedione). HEK293 cells were transduced to express human HSD3B2, and incubated medium containing P5 or DHEA. Depending on the incubation time with HSD3B2-expressing cells, the culture media progressively increased luciferase activities in CV-1 cells, transfected with the PR/AR expression vector and progesterone-/androgen-responsive reporter. Culture media from human and

FOOD SAFETY AND PUBLIC HEALTH INITIATIVES AT MBSTU

Mawlana Bhashani Science and Technology University, through the Department of Food Technology and Nutritional Science (FTNS), actively delivers regular community outreach and extension programmes aimed at improving public health, nutrition, and wellbeing. These initiatives engage faculty, students, and local stakeholders to promote awareness of food safety, balanced nutrition, maternal and child health, and personal hygiene. Special focus is given to supporting women, children, and disadvantaged groups in rural and peri-urban areas. Activities include workshops, health awareness campaigns, dietary counseling, and demonstrations on healthy cooking, safe food handling, and hygiene practices. Student volunteers play a key role in programme implementation, thereby enhancing experiential learning and community partnership. These efforts reflect our commitment to advancing SDG 3 by fostering healthier, more informed communities.

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ফুড টেকনোলজি এন্ড নিউট্রিশনাল সায়েন্স বিভাগ
মাওলানা ভাসানী বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
সন্তোষ, টাঙ্গাইল-১৯০২, মোবাইল : ০১৭৪৭-৩০৫৭৮৮
ফোন : ০৯২১-৬২৪০৯, ৫৫৩৯৯, ফ্যাক্স : ০৯২১-৫৫৪০০

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Date : 01/06/2024

To Whom It May Concern

Mawlana Bhashani Science and Technology University (MBSTU), as an institution which is committed to promote health and wellbeing within the local community through various outreach programmes and projects. The university actively encourages the participation of its faculties, departments, and students in community engagement initiatives that address public health, nutrition, hygiene, and related social issues. In line with this institutional commitment, the **Department of Food Technology and Nutritional Science (FTNS)** regularly organizes and participates in community outreach programmes aimed at improving awareness and practices related to nutrition, hygiene, food safety, and healthy living. Faculty members and students visit local communities, including disadvantaged and rural populations, to conduct educational campaigns, interactive sessions, and demonstrations that promote better health and wellbeing.

These activities not only contribute to enhancing public awareness and community health but also foster student volunteerism and experiential learning, enabling students to apply their academic knowledge to real-world problems and to develop a strong sense of social responsibility. Through these combined efforts, MBSTU as a whole plays an active role in advancing community health and wellbeing.

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বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়

Enclosure: *Some photographs demonstrating the outreach programmes conducted by the Department of FTNS in collaboration with local communities.*

COMMUNITY PARTNERSHIP IN SPORTS AND WELLBEING

Mawlana Bhashani Science and Technology University actively collaborates with nearby schools and colleges to promote physical fitness, sports participation, and overall wellbeing in the community, especially through the Department of Physical Education. Our university regularly shares its sports facilities with local educational institutions, providing access to modern equipment and well-maintained fields for practice and events. Schools and colleges frequently organize their annual sports competitions and inter-school tournaments at the university's central field, fostering community engagement and youth development. Joint training sessions and workshops are organized to enhance knowledge of sports science, health, and performance evaluation. The experts of the Department of Physical Education also mentor local coaches and physical education teachers, strengthening the region's sports ecosystem and supporting the SDG goal of ensuring healthy lives and promoting wellbeing for all ages.

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Ref: MBSTU/DPE/Rec.Copy/Pre./F/290 Date: 20/10/2024

To Whom It May Concern

Mawlana Bhashani Science and Technology University (MBSTU) as a body is committed to promoting physical fitness, sports participation, and overall well-being in the local community. In line with this commitment, the University, through its Department of Physical Education, actively shares its sports facilities, including well-maintained fields and courts, with nearby schools and colleges (e.g., Islamic University Boy's School, Islamic University Vocational School, Rani Din-Moni Primary School, and Mawlana Bhashani Ideal College, Tangail) for practice sessions, training programs, inter-school tournaments, and sports events.

The University encourages collaboration through joint training sessions, workshops, and seminars on sports science, health, and performance evaluation. Experts from the Physical Education Department also provide guidance and mentorship to local coaches and physical education teachers, supporting skill development and strengthening the regional sports ecosystem. Through these initiatives, the University fosters community ties, inspires young athletes to pursue higher education in sports science, and bridges the gap between academic research and real-world athletic practice. Overall, this partnership nurtures a culture of physical fitness, discipline, and lifelong learning across schools, colleges, and the University.

These initiatives reflect the University's policy to foster community engagement, youth development, and overall health and well-being, contributing to the broader Sustainable Development Goal (SDG) of ensuring healthy lives and promoting well-being for all ages. This collaborative effort also supports the development of a healthy generation prepared to compete globally.

20/10/2024

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ACCESSIBLE SEXUAL AND REPRODUCTIVE HEALTHCARE

Mawlana Bhashani Science and Technology University recognizes sexual and reproductive health as an essential component of overall well-being and provides dedicated services free of charge through its Medical Center, staffed by qualified health specialists. The center offers counseling, health education, menstrual hygiene management, and reproductive-care services to students and staff in a safe and confidential environment. Additionally, the Sabuj Chata Clinic (a government-initiated health center that provides sexual, reproductive and family planning services), is located within the university premises. It further supports these initiatives by delivering free family planning and reproductive health services. Together, these facilities and programs demonstrate the university's strong commitment to inclusive, accessible, and affordable health care for all members of the campus community.

SUSTAINABLE WELL-BEING: PSYCHOLOGICAL SUPPORT AND COUNSELING AT MBSTU

Mawlana Bhashani Science and Technology University demonstrates a strong commitment to student mental health and well-being through its dedicated Students' Welfare and Counseling Center (SWCC). The Center provides continuous psychological, emotional, and personal support to all students free of cost, ensuring accessibility, confidentiality, and inclusivity. Services include individual and group counseling, mental health awareness programs, stress management workshops, and guidance for personal development. The SWCC is supported by faculty counselors across departments and operates under clear policies and guidelines to maintain professional and ethical standards. By integrating mental health support into campus life, the University fosters holistic student development and contributes to sustainable social impact within the university community.

SMOKE-FREE POLICY: A COMMITMENT TO HEALTH

Mawlana Bhashani Science and Technology University maintains a 100% smoke-free campus policy, reflecting its strong commitment to promoting health, well-being, and environmental sustainability. The policy strictly prohibits smoking and the use of any tobacco products in all indoor and outdoor areas of the campus, including academic buildings, offices, residential halls, cafeterias, and open spaces. Awareness signage and campaigns are regularly implemented to inform and remind students, staff, and visitors about the health and environmental impacts of smoking. The University Medical Center and Students' Welfare and Counseling Center (SWCC) also provide free counseling and support services for individuals seeking assistance with smoking cessation. This policy aligns with national public health regulations and contributes to the Sustainable Development Goals (SDG 3: Good Health and Well-being) by fostering a clean, safe, and healthy learning environment for the university community.

The Smoke-Free Campus Policy of Mawlana Bhashani Science and Technology University was formally created and implemented in 2024, marking the University's institutional commitment to ensuring a clean, healthy, and tobacco-free environment. The policy was developed in accordance with the Smoking and Tobacco Products Usage (Control) Act of Bangladesh. The University displayed "No Smoking" signage at prominent locations to ensure awareness and compliance. This initiative reflects MBSTU's dedication to fostering long-term community health, well-being, and sustainability.

Mawlana Bhashani Science and Technology University (MBSTU)

Smoke-Free Policy

1. Purpose

The purpose of this policy is to protect the health and well-being of all students, faculty, staff, and visitors by ensuring a smoke-free environment across the university. MBSTU is committed to promoting a clean, healthy, and sustainable campus that supports the physical and mental wellness of the university community.

2. Scope

This policy applies to:

All students, faculty members, administrative and support staff, and visitors.

All university-owned, leased, or operated properties, including:

Academic buildings, laboratories, offices, and libraries

Halls of Residence, cafeterias, canteens, and sports facilities

Gardens, pathways, parking areas, and open spaces within campus boundaries

University vehicles and transportation services

3. Policy Statement

MBSTU is a 100% smoke-free campus.

The use, possession, or sale of tobacco products (including cigarettes, bidis, e-cigarettes, vaping devices, hookah, and smokeless tobacco) is strictly prohibited in all areas of the university campus. Smoking is not allowed within 100 feet of any campus building entrance, exit, or ventilation system. The university does not permit tobacco-related advertisements, sponsorships, or promotions in any form within its premises or at university events.

4. Definitions

Smoking: Inhaling, exhaling, burning, or carrying any lit tobacco or similar product, including e-cigarettes or vaping devices.

Tobacco products: Cigarettes, bidis, cigars, chewing tobacco, zarda, gul, khaini, and any other smokeless or electronic nicotine delivery systems.

University premises: All land, buildings, facilities, vehicles, and spaces under the university's jurisdiction.

5. Implementation and Responsibilities

The Registrar Office shall oversee implementation and ensure compliance.

The Proctor Office and Campus Security will monitor adherence to the policy.

The Directorate of Student Welfare and Counselling and the Medical Center will organize awareness campaigns on the harmful effects of smoking. All the departments/offices and the authorities of halls of residence are responsible for enforcing this policy within their respective areas.

6. Awareness and Education

Anti-tobacco education programs and campaigns shall be conducted regularly. "No Smoking" signs will be displayed prominently in all buildings and open spaces. Orientation programs for new students and staff will include information on this policy.

7. Support for Smoking Cessation

The university shall provide or refer individuals to counseling and cessation programs through:

The Directorate of Student Welfare and Counselling Centre, the University Medical Center, in Collaboration with local health organizations or NGOs specializing in tobacco cessation.

8. Enforcement and Penalties

Any individual found violating the policy shall be subject to disciplinary action according to university rules. Visitors who violate the policy may be asked to leave the premises. Persistent violations may result in further administrative actions as determined by university authorities.

9. Monitoring and Review

This policy will be reviewed every three years by a committee appointed by the Vice-Chancellor to ensure its effectiveness and compliance with national tobacco control laws. Feedback from students and staff will be considered in the review process.

Approved By: Proper Authority

12/11.2024

Md. Maksudur Rahman
Deputy Registrar
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COMMITMENT TO STAFF WELL-BEING

Mawlana Bhashani Science and Technology University is committed to promoting the mental health, emotional well-being, and overall personal development of its staff. The University provides free-of-charge professional psychological support and health guidance to its staff through its University Medical Center. Services include individual counseling, crisis support, health guidance, referrals to specialized mental health professionals, and occasional awareness programs. All services are accessible to every staff member, and confidentiality is strictly maintained. A photograph documenting an employee receiving counseling, along with a list of appointed health specialists and their duty roster, demonstrates the University's active engagement. These initiatives foster a supportive, healthy, and productive working environment, reflecting the University's commitment to sustainable staff well-being and institutional social impact.